

NIGERIA 2026 FEDERAL APPROPRIATION BUDGET

Prepared By:

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Tools Used: Tabula | Microsoft Excel | PGAdmin4 (PostgreSQL) | Microsoft Power BI | Microsoft Word | Microsoft PowerPoint

1. Project Overview

This project was commissioned by a leading Nigerian television station as the data backbone for a prime-time investigative report on the 2026 Federal Appropriation Budget. The target audience is everyday Nigerians, market traders, teachers, civil servants, parents, and young professionals, who pay taxes and deserve to understand, in plain language, where their money is going.

The analytical team was tasked with three deliverables: (1) a transparent record of data sources and cleaning decisions; (2) a set of analytical questions worth asking about the budget; and (3) a published data report that tells the story of the 2026 budget in a way any Nigerian can read and understand.

The total appropriation under analysis is ₦58,472,628,944,759, Fifty-Eight Trillion, Four Hundred and Seventy-Two Billion Naira, for the fiscal year ending 31 December 2026.

2. Data Source

A single primary source was used for this entire project:

Item	Detail
Document Name	2026 Federal Appropriation Bill
Issuing Authority	Budget Office of the Federation — Federal Ministry of Budget & Economic Planning (FMBEP)
Document Length	10 pages
Format	PDF (scanned structured layout)
Coverage	Full budget: revenues, expenditures, statutory transfers, debt service, capital allocations
Why Trusted	Official government document presented to the National Assembly. Primary source — not a media summary or secondhand account

3. Data Extraction Process (Tabula)

The 2026 Appropriation Bill was provided as a PDF, a format designed for printing, not analysis. The structured data inside it (budget lines, amounts, part labels) had to be extracted into a machine-readable format before any analysis could begin.

3.1 Tool Used: Tabula

Tabula is an open-source application specifically designed to extract tables from PDF documents. It works by detecting the grid structure of tables on each page and converting them into rows and columns that can be exported as CSV or Excel files.

3.2 Extraction Steps

Step	Action	Outcome
1	Opened the 2026 Appropriation Bill PDF in Tabula	All 10 pages loaded
2	Selected table regions on each page manually	Tabula detected budget line rows and amount columns
3	Ran extraction in "Lattice" mode (grid-based detection)	Preserved column boundaries correctly for structured budget tables
4	Exported the raw extraction as an Excel/CSV file	Raw file: tabula-2026_Appropriation_Bill_csv.xlsx
5	Opened the raw file in Microsoft Excel for inspection	Confirmed 22+ sheets of raw extracted data requiring cleaning

3.3 What Tabula Extracted vs. What It Missed

Tabula successfully extracted the majority of the budget schedule lines. However, several categories of content required manual attention after extraction:

Extraction Issue	Description	How Handled
Truncated text	Long line item names cut off mid-word (e.g. "AGRICULTURE AND FOO SECURITY")	Manually completed from PDF
Header row repetition	"SCHEDULE / 2026 BUDGET PROPOSAL" headers appeared as data rows	Deleted during cleaning
Sub-total rows	SUB-TOTAL: EXECUTIVE, TOTAL CRF CHARGES etc. mixed with line items	Removed — aggregates computed in SQL
Floating decimal	INEC amount extracted as 1013778401602.079956	Rounded to integer naira
Wrong numeric value	Development Levy recurrent extracted as 94,538,656,847	Corrected to 94,975,890,445 from PDF page 5
Carriage returns	_x000D_ characters embedded in text strings	Stripped during cleaning

4. Data Cleaning Decisions

All cleaning was performed in Microsoft Excel before the data was loaded into PostgreSQL. Every decision below is documented with a reason. Nothing was changed without justification.

4.1 What Was Kept

Every atomic budget line item from Parts A through Q of the Appropriation Bill Schedule was retained. This includes revenue lines, deductions, statutory transfers, debt service components, recurrent MDA allocations, pension lines, CRF charges, capital MDA allocations, capital supplementation lines, and financing items, 274 rows in total.

4.2 What Was Dropped

Dropped Item	Reason
Sub-total rows (e.g. SUB-TOTAL: EXECUTIVE = ₦9.5T)	These are derived aggregates. Recomputing them in SQL from atomic rows avoids double-counting and gives cleaner analysis.
Section header rows repeated by Tabula	Not data — they are formatting artefacts of the PDF extraction process.
Blank rows between schedule sections	No data content — visual spacing from the original PDF.
Signature Bonus / Renewals (₦0)	Zero-value line with no analytical content. Retained in dataset but excluded from positive-sum calculations.
Bank of Industry (BOI) dividend (₦0)	Zero-value line. Same treatment as above.

4.3 Columns Added During Cleaning

Four analyst-created columns were added to make the dataset ready for meaningful analysis. None of these exist in the original bill — they are the result of deliberate classification decisions made by the analyst team.

Column Added	What It Contains	Why It Was Added
expenditure_type	Revenue Statutory Transfer Debt Service Recurrent Capital Financing	Enables filtering by budget category — the most fundamental cut for any budget analysis
sector	13 thematic sectors e.g. Health, Security & Defence, Infrastructure	Groups MDAs and line items by policy purpose — essential for cross-sector comparison charts in Power BI
capital_nature	New Investment Carryover Obligation Loan-Funded (capital rows only)	Required to answer Q4 — without this classification the ₦23.2T capital headline cannot be decomposed
is_health_eligible	TRUE or FALSE boolean flag	Required to answer Q5 — identifies the 15 specific lines that count toward the Section 4 legal health compliance test

4.4 Negative Amounts

Six revenue rows carry negative amounts in the dataset. These are the Part B deduction rows (Oil & Gas Deductions, Non-Oil Deduction, FCT/Ecological Fund transfer) and the GOE Operating Surplus offset. Negative signs were preserved intentionally. They are correctly excluded from revenue totals using `amount_ngn > 0` filters in all views, ensuring they do not distort the aggregate revenue figure of ₦33.2 trillion.

5. Final Dataset Structure

The final analysis-ready dataset is stored as `nigeria_budget_2026_clean.csv` and loaded into the PostgreSQL table `budget_projections`.

5.1 Dataset Dimensions

274 Total Rows Individual budget line items	12 Columns Including 4 analyst-created	2026 Budget Year FY ending 31 Dec 2026	NGN Currency Nigerian Naira — integer
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5.2 Column Dictionary

Column Name	Data Type	Description
<code>budget_year</code>	INTEGER	Year of the budget (2026)
<code>currency</code>	VARCHAR(5)	Currency code (NGN)
<code>head_no</code>	VARCHAR(20)	Original head/line number from the Appropriation Bill
<code>line_item</code>	TEXT	Full name of the budget line item — cleaned and corrected
<code>budget_part</code>	VARCHAR(10)	Schedule part (PART A through PART Q)
<code>part_label</code>	TEXT	Descriptive label of the budget part
<code>expenditure_type</code>	VARCHAR(30)	Revenue Statutory Transfer Debt Service Recurrent Capital Financing
<code>sector</code>	VARCHAR(50)	Thematic sector assigned by analyst (13 sectors)
<code>sub_sector</code>	VARCHAR(80)	Granular sub-sector within the sector
<code>capital_nature</code>	VARCHAR(30)	Capital rows only: New Investment Carryover Obligation Loan-Funded
<code>is_health_eligible</code>	BOOLEAN	TRUE if line item counts toward the Section 4 health compliance test
<code>amount_ngn</code>	BIGINT	Appropriated amount in Nigerian Naira. Negative = deduction.

5.3 Expenditure Type Breakdown

Expenditure Type	Row Count	Description
Recurrent	125	Operational spending by MDAs, pensions, CRF charges
Capital	106	Infrastructure and development investment by MDAs
Revenue	22	All projected government income streams
Statutory Transfer	15	Mandatory transfers to agencies by law
Debt Service	3	Domestic debt, foreign debt, sinking fund
Financing	3	Borrowing and asset sales plugging the deficit

6. Data Loading into PostgreSQL via PGAdmin4

After cleaning in Excel, the CSV was loaded into a PostgreSQL database using PGAdmin4 — a graphical interface for managing PostgreSQL databases. The process used the COPY command to bulk-load all 274 rows in a single operation.

6.1 Why PostgreSQL?

PostgreSQL was chosen because: (1) it handles BIGINT amounts up to ₦58 trillion without rounding errors — unlike Excel which loses precision on very large numbers; (2) it supports complex CTEs and window functions needed for the analytical views; and (3) Power BI connects directly to PostgreSQL, allowing live queries rather than static snapshots.

6.2 Views Created for Power BI

16 named PostgreSQL views were created — one for each chart on the three dashboard pages. Power BI imports these views as datasets. This approach means the analytical logic lives in the database, not in Power BI DAX formulas, making it easier to audit, update, and explain.

Dashboard Page	Views Created	Purpose
Page 1 — Deficit & Debt Burden	6 views	Fiscal position, revenue sources, expenditure summary, debt composition, financing breakdown, kobo-per-naira
Page 2 — Capital Expenditure Illusion	5 views	Capital nature summary, carryover items, loan-funded projects, true capital waterfall, top MDA investment
Page 3 — Healthcare Compliance Test	5 views	Compliance test, health line items, health by type, sensitivity scenarios, health vs security comparison